Turbonator Tuners Guide

Robert Lloyd
BoostButton.com

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Section 1 - Fuel Tuning

1.1 - Wide-Open-Throttle

We'll start with WOT operation because it's the easiest to tune and the easiest to understand. The simplest situation for WOT is comes after the engine is at normal operating temperature. So, for this section, I'm going to assume that the engine is up to temperature. I'll go into the temperature compensation factors in a later section.

For Wide-open operation, the fuel is calculated from these values:

(WOTTBL_FuelFullThrottle + SparkScatterFuelStabilizationValue)* SumOfFuelModifiers - Factor_StartFuelDecayIntoRunFuel

WOTTBL is the Full-Throttle fuel PW table vs. MAP. The value in this table is an ideal PW based on MAP and a target A/F. This target A/F is hard-coded into the PW table – it's not a value that the computer uses for feedback or anything like that. It's just the value that Chrysler (or the tuner) chose when the table was created. Using CHeM or MP Tuner, you can see the effect of changing this target A/F on the PW

On the other hand, the spark scatter fuel stabilization value should only come into play at idle, and should be 0 at WOT. I'll cover this value in detail in the idle control section.

Also, the Factor_StartFuelDecayIntoRunFuel only comes into play during the first few seconds after starting the engine and also can be considered to be zero for most WOT applications. You definitely should not be firing the engine and going straight to WOT. But, if you did, this factor would affect the fuelling.

So, that really leaves only the SumOfFuelModifiers and the WOTTBL PW values for WOT fuel calculation. WOTTBL is the 16-bit base PW lookup from MAP, the SumOfFuelModifiers value is a fraction (or percentage) and is calculated as follows (for WOT operation, there are additional factors for P/T):

SumOfFuelModifiers = Cold Enrich factor * Cold Load enrich factor * Pump Eff

The cold enrich factor is calculated from the 3 cold start tables (A,B,C) and the time constants that phase them out. Once the engine is fully warmed up, the A and C factors go to 1 and so they will not have an effect on WOT fuel. So, we can ignore them (for now, we'll get back to them for cold enrichment).

The CurveB factor is a little bit different. For CurveB there is no decay factor for time – it is always active. At normal operating temperature, the factor goes to 1. However, CurveB is also multiplied by a factor (FCRCVB) at WOT. This value comes from a lookup table vs. RPM and the scale is 100-200%.

The Cold Load Enrichment factor only comes into play during warm-up.

The Pumping Efficiency table is always used.

So, at WOT, and under normal operating conditions (ie, post warm-up period), there are only 3 values that affect fuel:

WOTTBL (the main PW lookup) * PEFTBL (Pumping Efficiency) * CurveB * FCRCVB (the CurveB WOT Multiplier)

The only inputs to these tables are MAP, RPM, and Coolant temp.

WOT Tuning notes:

When tuning for WOT, typically you set the target A/F using the main fuel table (WOTTBL). CHeM or MP Tuner are very useful for helping you set these A/F values. If the actual WOT A/F is very far from this target, you should make adjustments with the PumpEff table.

I would advise caution when adjusting the CurveB or FCRCVB tables unless you are working with a highly modified cylinder head or intake system.

In setting up the Pumping Efficiency table, I would start with a stock table appropriate for your engine (is, use a 2.2 table for a 2.2 engine, 2.5 Masi, etc.). Generally speaking, cylinder head porting and/or high-lift cams tend to rotate the Pump Eff table counter-clockwise about the peak. So, the higher RPM range increases efficiency and the lower RPM range loses efficiency. It's also worth noting that Chrysler left the beginning and end of this table open. It's not realistic that the pumping efficiency is constant above 6000rpm. But, in setting the table up this way, they ensure that there is an excess of fuel. This helps keep cylinder temperatures and pressures down at these limits. This helps headgasket durability, but it kills power.

1.2.1 Part Throttle (Cruise)

Part throttle cruise is calculated similarly to WOT except that the adaptive memory and O2 feedback come into play. For P/T fuel:

(BasePWLookup + SparkScatterFuelStabilizationValue)* SumOfFuelModifiers - Factor_StartFuelDecayIntoRunFuel

Where the BasePWLookup is the lowest of these 2 values:

(MAPTBL_FuelPartThrottle * Timer_OpenLoopFraction) or WOTTBL_FuelFullThrottle.

Like WOT, the SparkScatterFuelStabilizationValue should be zero except at idle, and Factor_StartFuelDecayIntoRunFuel should be zero a few seconds after starting.

The 'SumOfFuelModifiers' value is calculated the same as for WOT with one exception: At part throttle, the adaptive memory value is included. The adaptive memory value can be +/- 25%. I'll show how the O2 controller works and how adaptive memory is calculated in the next chapter. For P/T operation, the 'SumOfFuelModifiers' value is calculated as:

SumOfFuelModifiers = Cold Enrich factor * Cold Load enrich factor * Pump Eff * 1 +/- ValueFromAdaptiveMemory

The Cold Enrich, Cold Load Enrich, and Pumping Efficiency factors are the same as for WOT. The adaptive memory value depends on the operating parameters of the engine and the O2 correction. There are 12 'cells' separated by RPM and MAP:

Auto Cal Factors Table
Closed throttle (idle) = cell 12, otherwise:

RPM ↑	8192	**1**	**3**	**5**	**7**	**9**	**11**
	R1	**0**	**2**	**4**	**6**	**8**	**10**
	0	MP1	MP2	MP3	MP4	MP5	14.7
	MAP →						

R1 and MP1 thru MP5 are all constants and can be changed by the tuner (if desired). Really, tuning these points should not normally be necessary.

The value stored in the cells comes from the O2 controller. Basically, if the O2 does not toggle soon enough, then the adaptive cell gets updated with a new value. The value is signed and can be +/- 25%.

The adaptive memory value is a RAM variable calculated by the O2 Ramp Controller. So, these values cannot be tuned directly.

1.2.2 O2 Controller Ramp Calibration

The O2 sensor is monitored ONLY for "rich/lean". A change from one to the other is a "toggle". The system constantly toggles between rich and lean so that the overall average is around stoichiometric.

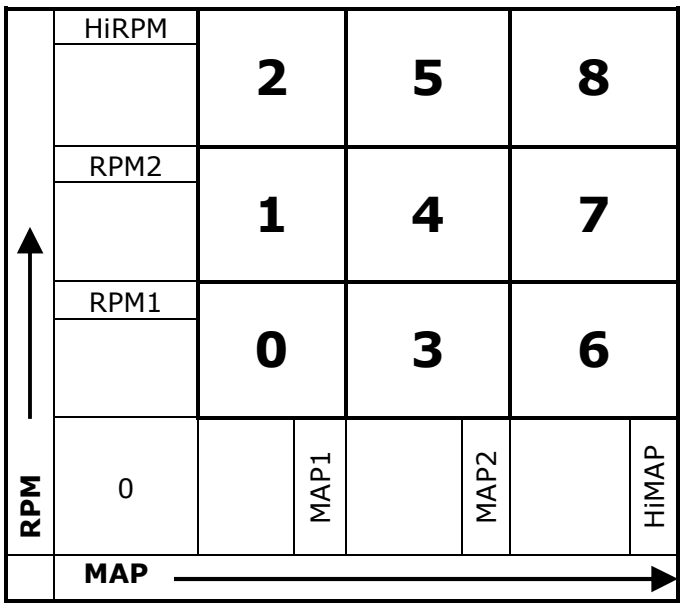
This system divides a range of RPMs and MAP readings into "cells". The number of "kicks", or increases/decreases in fuel required to toggle the O2 reading, is different for each of these cells. This allows some fairly powerful flexibility in tuning. The only time you might want to adjust these values is to speed or slow the adjustment being performed.

Each cell has a 'kick' value. This is the amount of change applied to the running fuel until a rich/lean toggle occurs. The net result is the familiar rich/lean flashing that your O2 sensor gauge shows you.

The 'limits' are the maximum number of kicks that may be applied to each cell. If the O2 sensor fails to toggle after reaching the limit for more than a few sec, either the AutoCal cell gets changed (in the case of the secondary ramp) or a code for the O2 Sensor is stored (in the case of the primary ramp).

Primary Ramp vs. Secondary Ramp:
The Primary ramp is responsible only for the normal O2 controller feedback and setting the O2 sensor codes. The Secondary ramp is also responsible for O2 feedback, and also for updating the adaptive memory cells.

O2 Controller Cells
Closed Throttle (Idle) = Cell 9, otherwise:



For each cell, there are 2 numbers. The first number is "rich" value, the second is "lean" value. If the O2 sensor is returning a lean value, "rich" kicks are added. If the O2 sensor is returning a rich value, "lean" kicks are added.

KICK = (factor * 512) - 512
example: 5% -> 1.05 * 512 - 512 = 25.6 (dec) = 0x19 ; each bit is ~ 0.2%

P/T Tuning Notes:
Turning Off AutoCal (Adaptive memory):
Add this sub-section…

Tuning for lean cruise:
Add this sub-section…